

Microbes & cells — overview

Immune cells, pathogens and body cells side by side

Variant: basics (~30 entries)

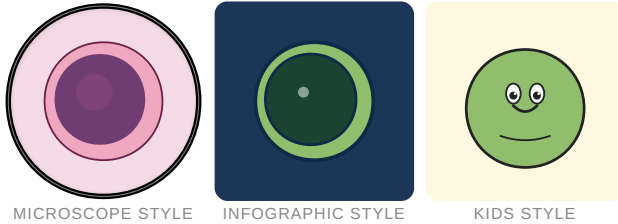
Contents

- | | |
|--|--|
| 1. 1. Immune cells — Lymphocytes | 6. 6. Hematopoietic lineage |
| 2. 2. Immune cells — Phagocytes & Granulocytes | 7. 7. Mesenchymal lineage |
| 3. 3. Pathogens | 8. 8. Neural lineage |
| 4. 4. Well-known bacteria | 9. 9. Epithelial lineage |
| 5. 5. Well-known viruses & other pathogens | 10. 10. Endothelial lineage |
| | 11. 11. Induced pluripotent stem cells (iPS) |

Where a real microscope photograph is available, it is shown; otherwise a stylised microscope-style sketch. The infographic and kids styles are hand-drawn SVGs.

Lymphocytes are small white blood cells with a large round nucleus. They mature in bone marrow (B, NK) or the thymus (T), circulate through blood and lymph, and gather in lymph nodes, spleen and mucosal tissues.

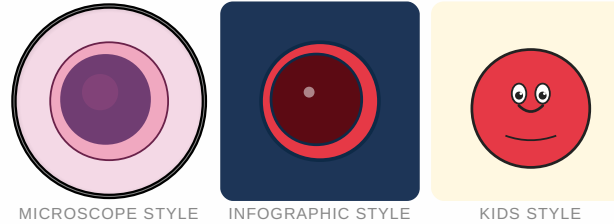
Helper T cell (CD4)



FUNCTION: Conductor of the immune response: recognises antigen fragments shown by other cells and releases messengers that switch other cells on.

DEPENDS ON / TARGETS: Needs antigen-presenting cells (dendritic cells, macrophages). Activates B cells and CD8 killer cells.

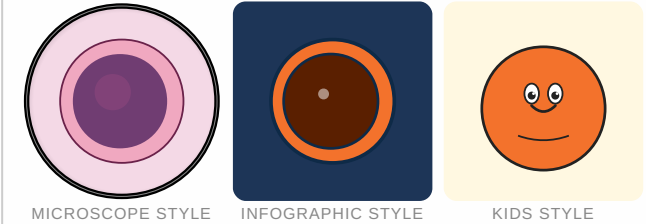
Cytotoxic T cell (CD8)



FUNCTION: Killer cell: spots infected or tumour cells by their MHC-I signal and kills them precisely with perforin and granzymes.

DEPENDS ON / TARGETS: Primed by dendritic cells, often with help from CD4 T cells. Targets virus-infected body cells and tumour cells.

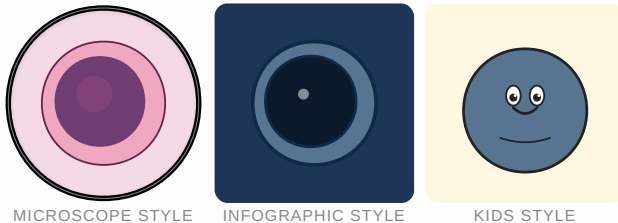
B cell



FUNCTION: Antibody factory: binds antigen with its B-cell receptor and matures into plasma cells that pump out tailored antibodies.

DEPENDS ON / TARGETS: Usually needs help from CD4 T cells to switch into high gear. Targets free bacteria, virus particles and toxins in blood and tissue.

Natural killer cell (NK)

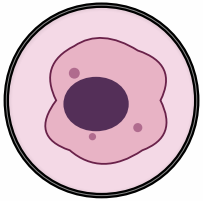


FUNCTION: Innate-immune bouncer: kills cells that have lost their MHC-I 'ID badge' — typical for virus infection or cancer.

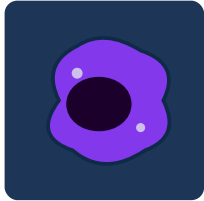
DEPENDS ON / TARGETS: Revved up by interferons and IL-12. Kills virus-infected and tumour cells without prior priming.

Phagocytes and granulocytes come from the myeloid lineage in bone marrow. They engulf, digest and neutralise invaders, patrolling blood, tissues and entry points like skin, lung and gut.

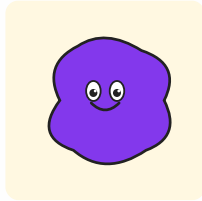
Macrophage



MICROSCOPE STYLE



INFOGRAPHIC STYLE

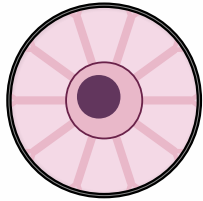


KIDS STYLE

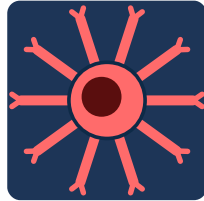
FUNCTION: 'Big eater': engulfs bacteria, dead cells and debris, presents fragments to T cells and calls reinforcements with cytokines.

DEPENDS ON / TARGETS: Differentiates from blood monocytes. Phagocytoses bacteria, fungi, cancer cells and cleans up wounds.

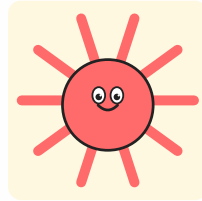
Dendritic cell



MICROSCOPE STYLE



INFOGRAPHIC STYLE

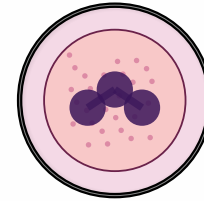


KIDS STYLE

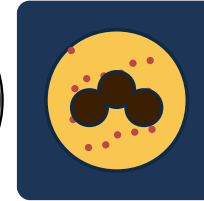
FUNCTION: Scout and storyteller: gathers antigen in tissues, travels to lymph nodes and shows it to T cells — kicks off the adaptive response.

DEPENDS ON / TARGETS: Bridges innate and adaptive immunity. Activates helper T cells and CD8 killer cells.

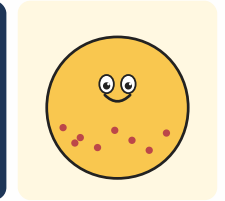
Neutrophil



MICROSCOPE STYLE



INFOGRAPHIC STYLE



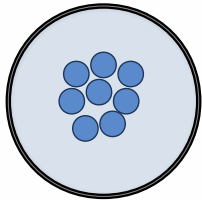
KIDS STYLE

FUNCTION: First responder: arrives first at sites of inflammation, eats bacteria, dumps antimicrobials and even casts sticky DNA nets (NETs).

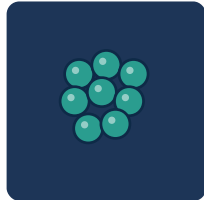
DEPENDS ON / TARGETS: Chemokine-guided. Targets bacteria and fungi; lives only hours to a few days.

Pathogens are anything that can cause disease: bacteria, viruses, fungi, parasites and prions. They live in air, water, soil, animals and on our skin — most microbes are harmless or beneficial, a few are highly dangerous.

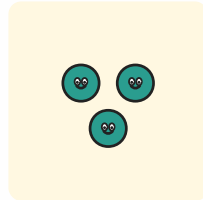
Cocci (round bacteria)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

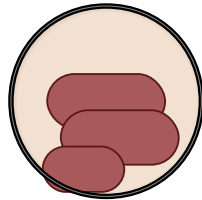


KIDS STYLE

FUNCTION: Round bacteria, often in clusters or chains (Staphylococci, Streptococci). Some are harmless guests, others cause pneumonia, skin or throat infections.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and macrophages; antibodies help tag them.

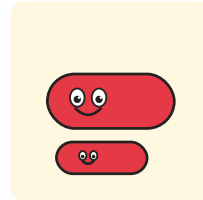
Rod bacteria



MICROSCOPE STYLE



INFOGRAPHIC STYLE

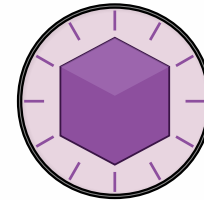


KIDS STYLE

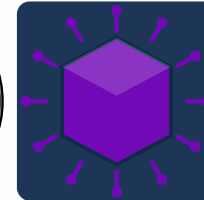
FUNCTION: Elongated bacteria like E. coli, Salmonella, tuberculosis bacilli. Multiply fast and can release toxins.

DEPENDS ON / TARGETS: Phagocytosed; intracellular species (TB) need T-cell help to be cleared.

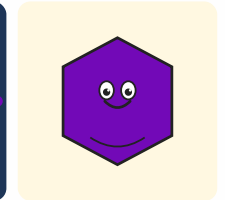
Virus



MICROSCOPE STYLE



INFOGRAPHIC STYLE

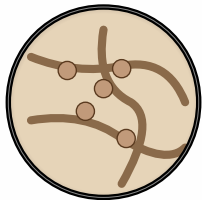


KIDS STYLE

FUNCTION: Not really alive: just genetic material in a shell. Sneaks into body cells and forces them to build new virus particles.

DEPENDS ON / TARGETS: Hunted by CD8 T cells and NK cells; antibodies bind free virus particles.

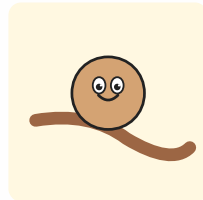
Fungus



MICROSCOPE STYLE



INFOGRAPHIC STYLE

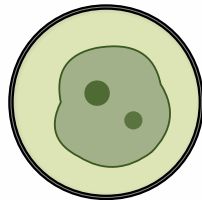


KIDS STYLE

FUNCTION: Eukaryotic microbes — yeasts or filaments (hyphae). Examples: Candida in mouth/gut, Aspergillus in lungs.

DEPENDS ON / TARGETS: Controlled by neutrophils, macrophages and Th17 cells.

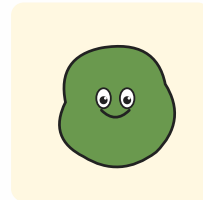
Parasite



MICROSCOPE STYLE



INFOGRAPHIC STYLE

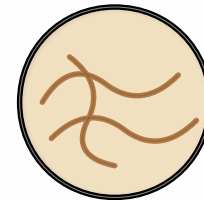


KIDS STYLE

FUNCTION: From single-celled (Plasmodium → malaria) to worms. Lives in or on its host and damages it in the process.

DEPENDS ON / TARGETS: Often fought with IgE, eosinophils and mast cells; antibodies and Th2 responses are central.

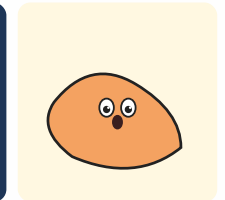
Prion



MICROSCOPE STYLE



INFOGRAPHIC STYLE



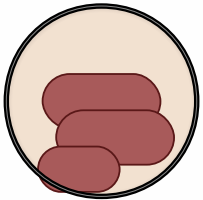
KIDS STYLE

FUNCTION: Not alive, not a virus — a misfolded protein that forces other proteins to misfold the same way. Causes diseases like Creutzfeldt-Jakob, BSE.

DEPENDS ON / TARGETS: Largely invisible to the immune system; no antibody defence.

Bacteria are single-celled organisms without a true nucleus (prokaryotes). They live everywhere — in soil and water, on skin and in the gut. Most are harmless or beneficial; a few cause millions of cases of disease worldwide every year.

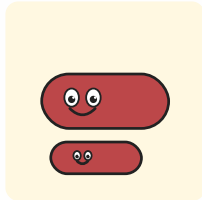
Mycobacterium tuberculosis (TB)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

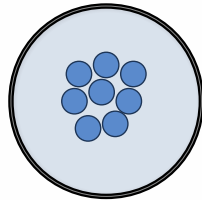


KIDS STYLE

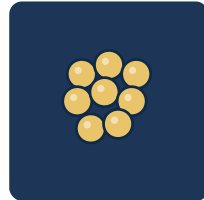
FUNCTION: Waxy-coated rod that hides inside macrophages and can lurk in the lungs for years. Causes tuberculosis — still one of the deadliest infections worldwide.

DEPENDS ON / TARGETS: Airborne transmission. Contained by macrophages and CD4 T cells; without them the bacterium persists.

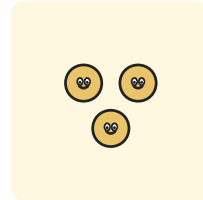
Staphylococcus aureus (MRSA)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

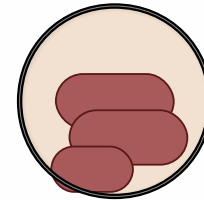


KIDS STYLE

FUNCTION: Grape-cluster cocci. Often harmless on skin and in the nose, but can cause wound infections, abscesses and sepsis. MRSA strains resist many antibiotics.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and tagged by antibodies. Hospital hygiene is key against resistance spread.

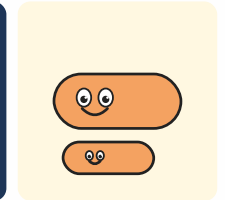
Escherichia coli



MICROSCOPE STYLE



INFOGRAPHIC STYLE

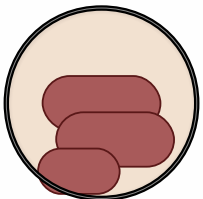


KIDS STYLE

FUNCTION: Gut rod bacterium. Most strains are harmless or useful; some (e.g. EHEC) produce toxins and cause severe diarrhoea and kidney failure.

DEPENDS ON / TARGETS: Spread by faecal-oral route and contaminated food. Controlled in the gut by IgA antibodies and the mucus barrier.

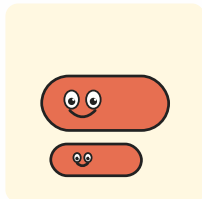
Salmonella enterica



MICROSCOPE STYLE



INFOGRAPHIC STYLE



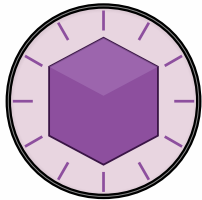
KIDS STYLE

FUNCTION: Rod bacterium from many animals' guts. Picked up via contaminated eggs, poultry or water, it causes diarrhoea (salmonellosis) or typhoid fever.

DEPENDS ON / TARGETS: Cleared by neutrophils, macrophages and the gut epithelium; Th1 responses are central to elimination.

Viruses are genetic material in a coat and need a host cell to replicate. Fungi and parasites, by contrast, are organisms in their own right. The pathogens shown here infect humans worldwide, ranging from harmless colds to severe pandemics.

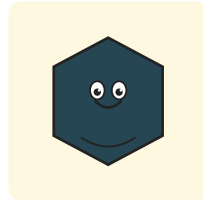
Influenza virus (flu)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

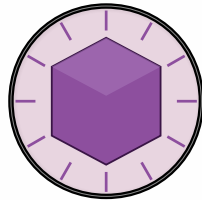


KIDS STYLE

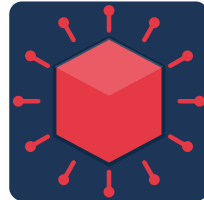
FUNCTION: Respiratory RNA virus with a segmented genome. Mutates constantly, so vaccines are reformulated each year. Causes seasonal waves and occasional pandemics.

DEPENDS ON / TARGETS: Spread by droplets. CD8 T cells and antibodies clear infected cells and free virus particles.

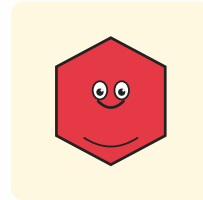
SARS-CoV-2 (COVID-19)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

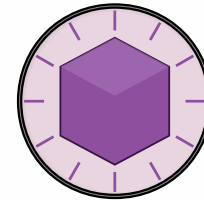


KIDS STYLE

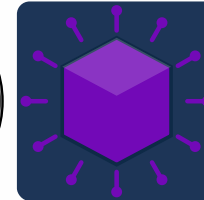
FUNCTION: Coronavirus with a 'crown' of spike proteins. Binds ACE2 receptors in the airways and triggered the global COVID-19 pandemic in 2020.

DEPENDS ON / TARGETS: Spreads via aerosols. mRNA and vector vaccines train antibodies and T cells against the spike protein.

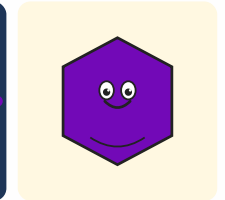
HIV



MICROSCOPE STYLE



INFOGRAPHIC STYLE

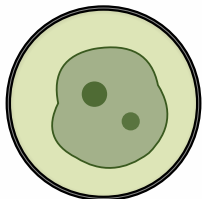


KIDS STYLE

FUNCTION: Retrovirus that targets CD4 helper T cells and integrates its genome into human DNA. Untreated, it destroys the immune system (AIDS).

DEPENDS ON / TARGETS: Spread via blood, sexual contact and mother-to-child. Antiretroviral therapy (ART) keeps the virus suppressed long-term.

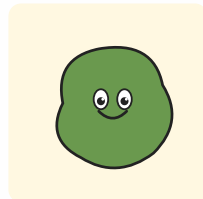
Plasmodium (malaria)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

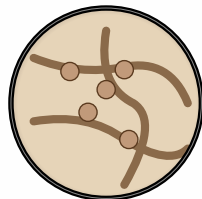


KIDS STYLE

FUNCTION: Single-celled parasite spread by Anopheles mosquitoes. First infects the liver, then red blood cells, triggering cycles of fever. Kills over 500,000 people per year worldwide.

DEPENDS ON / TARGETS: Treated with antimalarial drugs; new vaccines (RTS,S, R21) now being rolled out in endemic regions.

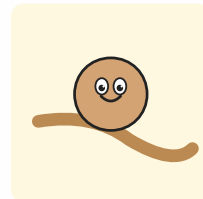
Candida albicans



MICROSCOPE STYLE



INFOGRAPHIC STYLE



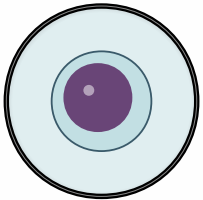
KIDS STYLE

FUNCTION: Yeast that lives harmlessly in many people's mouths, gut and genital tract. With a weakened immune system or after antibiotics it can spread and cause serious infections.

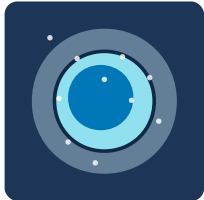
DEPENDS ON / TARGETS: Kept in check by Th17 cells, neutrophils and the mucosal microbiome.

All blood cells arise in bone marrow from a single hematopoietic stem cell. Through multiple maturation steps they become red blood cells, platelets, granulocytes, monocytes and lymphocytes — around 200 billion new cells per person per day.

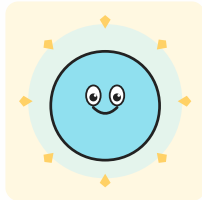
Hematopoietic stem cell (HSC)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

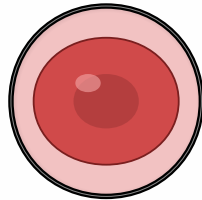


KIDS STYLE

FUNCTION: Mother of all blood cells. Lives in bone marrow, divides rarely, self-renews and gives rise to every blood lineage.

DEPENDS ON / TARGETS: Needs niche cells in bone marrow (osteoblasts, stromal cells) and growth factors.

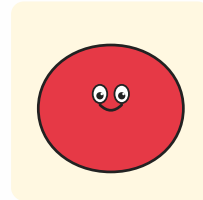
Erythrocyte (red blood cell)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

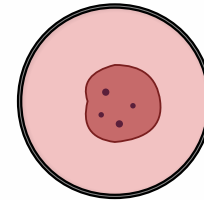


KIDS STYLE

FUNCTION: Oxygen carrier: no nucleus, packed with haemoglobin. Lives about 120 days and circulates millions of times.

DEPENDS ON / TARGETS: Production driven by the hormone erythropoietin from the kidney. Delivers oxygen to every body cell.

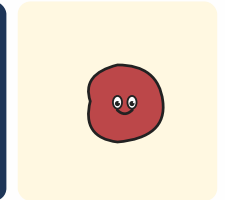
Thrombocyte (platelet)



MICROSCOPE STYLE



INFOGRAPHIC STYLE



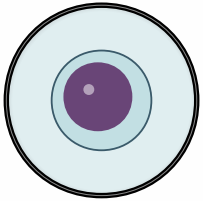
KIDS STYLE

FUNCTION: Tiny anucleate fragments. Plug wounds, form the first clot and trigger the coagulation cascade.

DEPENDS ON / TARGETS: Originate from megakaryocytes. Cooperate closely with coagulation factors and endothelial cells.

Mesenchymal cells arise from the middle germ layer (mesoderm) and form the body's supporting tissues: bone, cartilage, tendons, skeletal and smooth muscle, plus fat and connective tissue.

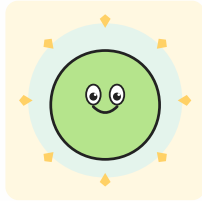
Mesenchymal stem cell (MSC)



MICROSCOPE STYLE



INFOGRAPHIC STYLE



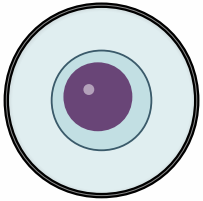
KIDS STYLE

FUNCTION: Reserve cell for supporting tissues. Can turn into bone, cartilage, muscle, fat or connective-tissue cells.

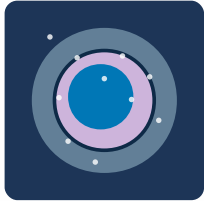
DEPENDS ON / TARGETS: Lives in bone marrow, fat tissue and umbilical cord. Supplies precursors on demand.

Neurons and glia arise from the neural tube (ectoderm) and build the entire nervous system — brain, spinal cord and the peripheral nerves that supply muscles, skin and organs.

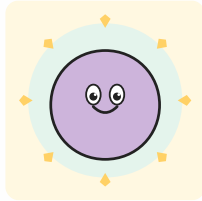
Neural stem cell (NSC)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

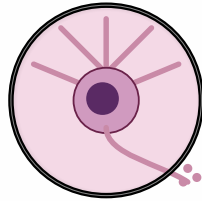


KIDS STYLE

FUNCTION: Precursor of all neural cell types. In the adult brain only active in a few niches (e.g. hippocampus).

DEPENDS ON / TARGETS: Responds to growth factors (EGF, FGF) and to tissue cues.

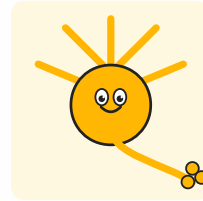
Neuron



MICROSCOPE STYLE



INFOGRAPHIC STYLE

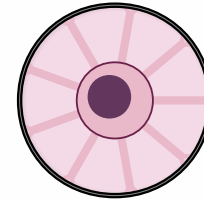


KIDS STYLE

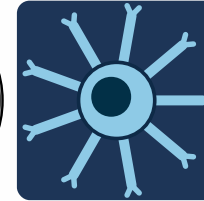
FUNCTION: Compute-and-signal cell: collects inputs through dendrites, sums them up and fires electrical impulses down its axon.

DEPENDS ON / TARGETS: Depends on astrocytes (support) and oligodendrocytes (myelin). Communicates with other neurons through synapses.

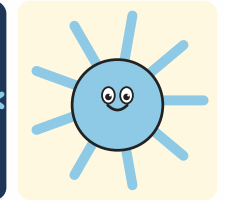
Astrocyte



MICROSCOPE STYLE



INFOGRAPHIC STYLE



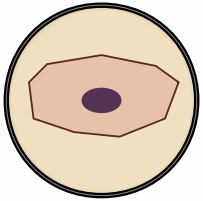
KIDS STYLE

FUNCTION: Star-shaped support cell. Feeds neurons, balances ions and neurotransmitters, helps form parts of the blood-brain barrier.

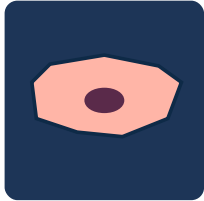
DEPENDS ON / TARGETS: Supports neurons and regulates local blood flow.

Epithelial cells line every inner and outer surface of the body: skin, the mucosa of mouth, GI tract and lung, plus glands. They form the body's first mechanical and biochemical barrier against the environment.

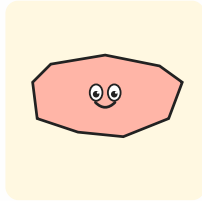
Keratinocyte (skin cell)



MICROSCOPE STYLE



INFOGRAPHIC STYLE

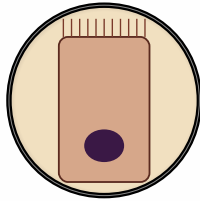


KIDS STYLE

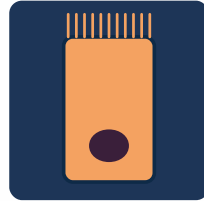
FUNCTION: Building block of the outer skin. Lays down keratin, cornifies upward and forms a waterproof, robust barrier.

DEPENDS ON / TARGETS: Renewed from stem cells in the basal layer. Cooperates with melanocytes and Langerhans cells.

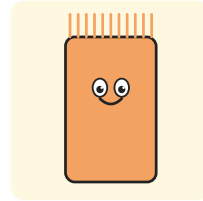
Enterocyte (gut cell)



MICROSCOPE STYLE



INFOGRAPHIC STYLE



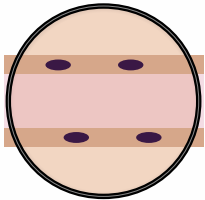
KIDS STYLE

FUNCTION: Tall column-shaped cell with a dense brush border (microvilli). Absorbs nutrients from the gut and passes them on.

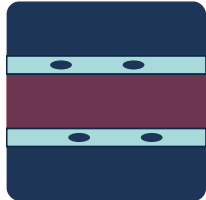
DEPENDS ON / TARGETS: Replaced every few days by intestinal stem cells. Forms a barrier against gut microbes.

Endothelial cells form a single layer that lines the inside of every blood and lymph vessel, from heart to the smallest capillary. They regulate blood flow, clotting and the passage of immune cells out of the bloodstream into tissues.

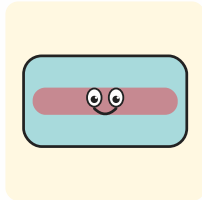
Endothelial cell (continuous)



MICROSCOPE STYLE



INFOGRAPHIC STYLE



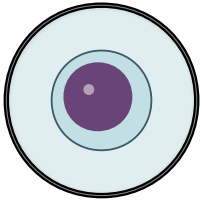
KIDS STYLE

FUNCTION: Flat cobblestone that lines most vessels. Regulates what diffuses between blood and tissue.

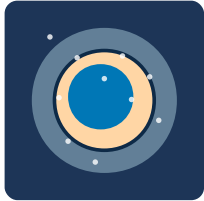
DEPENDS ON / TARGETS: Responds to blood-flow shear and hormones (e.g. histamine).

Induced pluripotent stem cells are made in the lab: body cells (e.g. skin cells) are reprogrammed with defined factors back into an embryonic-like stem-cell state and can then differentiate into almost any cell type.

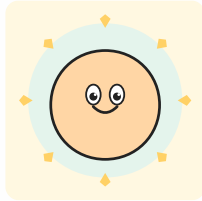
iPS cell



MICROSCOPE STYLE



INFOGRAPHIC STYLE



KIDS STYLE

FUNCTION: A body cell (e.g. skin) reset by a handful of transcription factors into an embryonic-like state — can develop into almost any cell type.

DEPENDS ON / TARGETS: Differentiated in vitro into a target lineage with specific growth factors.

Image credits

All real microscope images come from open sources (Wikimedia Commons, CDC PHIL, NIAID and similar) under free licences.

Helper T cell (CD4) — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:Healthy_Human_T_Cell.jpg

B cell — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:B_lymphocyte.jpg

Natural killer cell (NK) — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:NK_cell.jpg

Macrophage — Wikimedia Commons contributors CC BY-SA 4.0 <https://commons.wikimedia.org/wiki/File:Macrophage.jpg>

Dendritic cell — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:Dendritic_cell_revealed.jpg

Neutrophil — Volker Brinkmann / PLoS Pathogens CC BY 2.5 https://commons.wikimedia.org/wiki/File:Neutrophil_with_anthrax_copy.jpg

Cocci (round bacteria) — CDC PHIL Public domain (US Government work) https://commons.wikimedia.org/wiki/File:Streptococcus_pneumoniae.jpg

Rod bacteria — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:EscherichiaColi_NIAID.jpg

Virus — NIAID CC BY 2.0 [https://commons.wikimedia.org/wiki/File:Novel_Coronavirus_SARS-CoV-2_\(49531042482\).jpg](https://commons.wikimedia.org/wiki/File:Novel_Coronavirus_SARS-CoV-2_(49531042482).jpg)

Fungus — CDC PHIL Public domain (US Government work) https://commons.wikimedia.org/wiki/File:Candida_albicans_PHIL_3192_lores.jpg

Parasite — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Toxoplasma_gondii.jpg

Mycobacterium tuberculosis (TB) — CDC PHIL Public domain (US Government work) https://commons.wikimedia.org/wiki/File:Mycobacterium_tuberculosis_Ziehl-Neelsen_stain_02.jpg

Staphylococcus aureus (MRSA) — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:Staphylococcus_aureus_VISA_2.jpg

Escherichia coli — NIAID CC BY 2.0 https://commons.wikimedia.org/wiki/File:EscherichiaColi_NIAID.jpg

Salmonella enterica — NIAID CC BY 2.0 <https://commons.wikimedia.org/wiki/File:SalmonellaNIAID.jpg>

Influenza virus (flu) — CDC PHIL Public domain (US Government work) https://commons.wikimedia.org/wiki/File:EM_of_influenza_virus.png

HIV — CDC PHIL Public domain (US Government work) <https://commons.wikimedia.org/wiki/File:HIV-budding-Color.jpg>

Plasmodium (malaria) — CDC PHIL Public domain (US Government work) https://commons.wikimedia.org/wiki/File:Plasmodium_falciparum_01.png

Erythrocyte (red blood cell) — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Red_blood_cells.jpg

Thrombocyte (platelet) — Wikimedia Commons contributors CC BY-SA 4.0 <https://commons.wikimedia.org/wiki/File:Platelets2.JPG>

Neuron — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:GFP_Neuron.jpg

Astrocyte — Wikimedia Commons contributors CC BY-SA 4.0 <https://commons.wikimedia.org/wiki/File:Astrocyte5.jpg>

Keratinocyte (skin cell) — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Keratinocytes_in_culture.jpg

Enterocyte (gut cell) — Wikimedia Commons contributors CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Small_intestinal_villi.jpg

Endothelial cell (continuous) — Wikimedia Commons contributors CC BY-SA 4.0 <https://commons.wikimedia.org/wiki/File:HUVEC.jpg>